QQ-C-571B
August 27, 1974
SUPERSEDING
Fed. Spec. QQ-C-571A
April 17, 1967

FEDERAL SPECIFICATION

COPPER, PHOSPHOR (ALLOYING ADDITIVE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two phosphor copper alloys.

1.2 Classification. Phosphor copper shall be furnished in the following alloys and forms, as specified (see 6.2):

Alloys - A - 14 percent phosphorus
B - 10 percent phosphorus

Forms - Ingots, notched slab
Shot

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-585 - Boxes, Wood, Wirebound.

FSC 9630

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PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-1806 - Barrels and Kegs, Wood, Slack.
PPP-D-705 - Drum: Metal Shipping, Steel, (Over 12 and under 55 Gallon).
PPP-D-723 - Drums, Fiber.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Test Method Std. No. 151 - Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Office in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads (40" x 48" 4-Way Partial and 4-Way Pallets)

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standard:

B 52 - Phosphor Copper

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. Phosphor copper shall be made using copper with or without the addition of clean scrap resulting from the manufacture of phosphor copper of similar composition.

3.2 Chemical composition. The chemical composition of phosphor copper shall be as shown in table I.

Table I. Chemical composition, percent

Alloy	Phosphorus, min.	Phosphorus plus copper, min.	Iron, max.
A	14	99.75	0.15
B	10	99.75	.15

3.3 Form. Unless otherwise specified (see 6.2), phosphor copper shall be furnished in the form of notched slabs or ingots weighing from 15 to 20 pounds. When specified (see 6.2), material shall be furnished in the form of shot.

3.4 Identification marking. The manufacturer's name or brand and melt number shall be legibly and indelibly marked on each slab, ingot or container, by which the manufacturer and the heat may be identified.

3.5 Workmanship. All forms of phosphor copper shall be commercially clean and free from deleterious foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Lot. A lot shall consist of all phosphor copper of the same alloy and form submitted for inspection at one time.

4.3 Sampling.

4.3.1 For chemical analysis. A sample for chemical analysis shall be taken from each lot. It shall consist of not less than two ounces of material removed from the samples selected in accordance with ASTM B 52.

4.3.2 For visual examination. From each lot of material, samples shall be selected for visual examination in accordance with MIL-STD-105 at inspection level I. The acceptable quality level (AQL) shall be 2.5 percent defective.

4.4 Examination.

4.4.1 Visual. Each sample selected in accordance with 4.3.2 shall be subjected to surface examination for workmanship requirements.

4.4.2 Preparation for shipment. Examination of the preservation, packaging, packing, and marking for shipment shall be made for conformance with the requirements of section 5.

4.5 Test methods.

4.5.1 Chemical analysis. The samples selected in accordance with 4.3.1 shall be analyzed in accordance with method 111 or 112 of Fed. Test Method Std. No. 151. In case of dispute, referee analysis shall be by method 111.

4.6 Rejection.

4.6.1 Examination defects. Any sample unit having one or more defects shall be rejected. If the number of nonconforming sample units in the sample exceeds the acceptance number specified in 4.3.2 for that sample size, the entire lot shall be rejected subject to the provisions of the section on "Acceptance and Rejection" of MIL-STD-105.

4.6.2 Test failures. A lot shall be rejected for failure to meet any one of the test requirements when tested in accordance with 4.5, subject to the provisions of the section on "Rejection and Retests" of Fed. Test Method Std. No. 151.

5. PREPARATION FOR DELIVERY

5.1 Packing (see 6.2).

5.1.1 Level A.

5.1.1.1 Slabs and ingots. Unless otherwise specified (see 6.2), slabs and ingots shall be (1) packed separately in a box conforming to PPP-B-601, overseas type; or PPP-B-621, class 2; or PPP-B-585, class 3; or (2) packed in

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fiber drums conforming to PPP-D-723, type II, or in metal drums conforming to PPP-D-705. The gross weight of each box shall not exceed 550 lbs. The boxes shall be strapped in accordance with the appendix to the applicable box specification. When specified (see 6.2), slabs and ingots shall be prepared for shipment on pallets conforming to MIL-STD-147.

5.1.1.2 Shot. Shot of like size and description shall be packed, in quantities not to exceed 500 pounds, in fiber drums conforming to PPP-D-723, type II; in metal drums conforming to PPP-D-705; or in wood barrels conforming to PPP-B-1806.

5.1.2 Level B.

5.1.2.1 Slabs and ingots. Slabs and ingots shall be packed as specified in 5.1.1.1 except the containers shall be domestic type and class and the gross weight shall not exceed the weight limitations of the applicable box or drum specification.

5.1.2.2 Shot. Shot of like size and description shall be packed as specified in 5.1.1.2 except that fiber drums may be type I.

5.1.3 Level C. The products shall be separated by alloy and form and packed in accordance with the manufacturer's standard practice into containers of a type and size commonly used for the purpose, in such a manner as to insure acceptance by carrier for transportation at the lowest rate applicable and to afford maximum protection from normal hazards of transportation.

5.2 Marking (see 6.2 and 6.3).

5.2.1 Civil agencies. In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Phosphor copper is used as a vehicle for the addition of phosphorus in the melting and preparation of copper alloys.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Alloy and form required (see 1.2, 3.2, and 3.3).
(c) Size of ingots (see 3.3).
(d) Selection of applicable level of packing required (see 5.1).
(e) When Level A packing other than as specified is required (see 5.1.1.1).

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6.3 The requirements for product identification marking (see 3.4) and for packing and marking for shipment (see 5.1 and 5.2) specified herein apply to direct shipment for Government activities and apply also, where specified, to contracts or orders between the manufacturer and the Government prime contractor.

6.4 Related specification. This specification covers the same material as ASTM Specification B 52.

MILITARY CUSTODIANS:

Army - MR
Navy - OS
Air Force - None

Review activities:

Army - MR, WC, MU
Navy - OS
Air Force - None

User activities:

Army - None
Navy - SH
Air Force - None

Preparing activity:

Army - MR

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 25 cents each.

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